Perspective Homography

A Robot's Journey from Map to Reality

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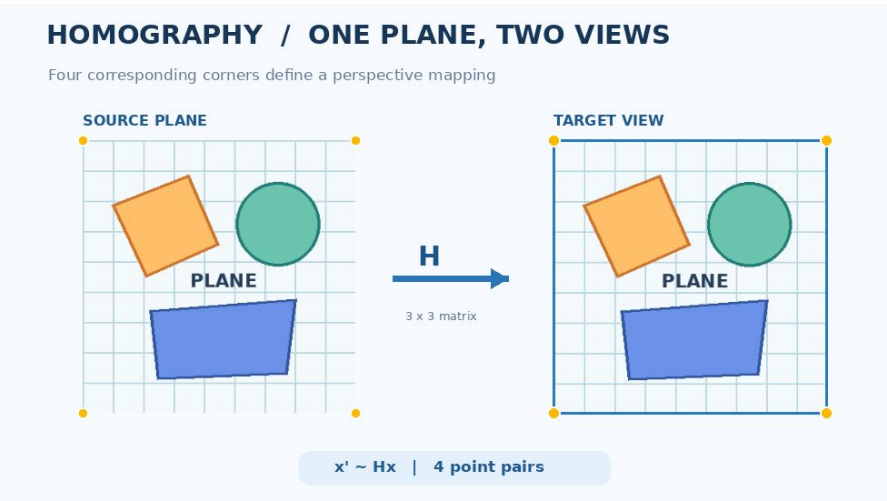
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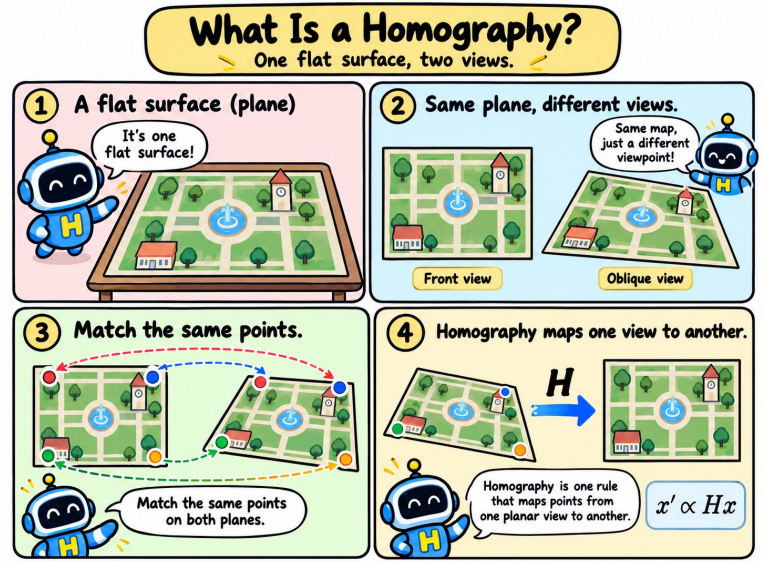


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Core Concepts

- **Definition:** A homography is a **projective transformation** between two planes.
- **Equation:** $x' \propto Hx$
  H: 3x3 homography matrix
- **Applied on: planar scenes/objects** or two views of the same plane.
- **Minimum requirement:** At least **4 point correspondences** are needed to estimate **H**.
- **Key Property:** Homography preserves **straight lines and collinearity**, but generally does **not preserve lengths, angles, or parallelism**.
- **Common Applications: Perspective rectification, panorama stitching, planar tracking, and pose estimation.**





OVERVIEW

How Homography Works

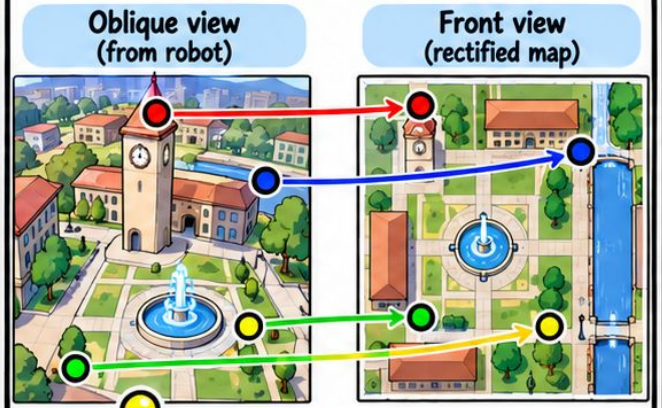
1. Understand the Perspective



A flat plane looks distorted from a different viewpoint.

VIEWPOINT

2. Match Corresponding Points



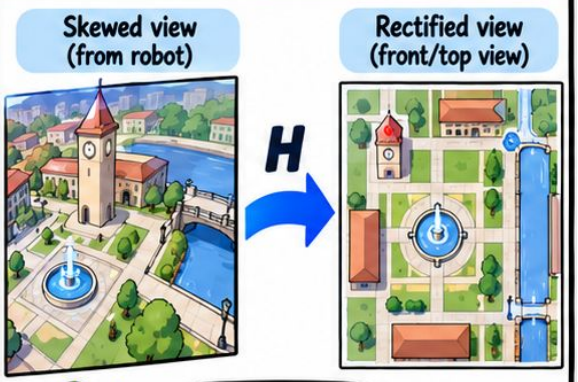


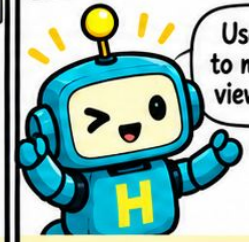
Match the same physical points in both views.

$x' \sim Hx$

MATCHING

3. Rectify the View





Use homography to map one planar view into another.

Works best for one plane.



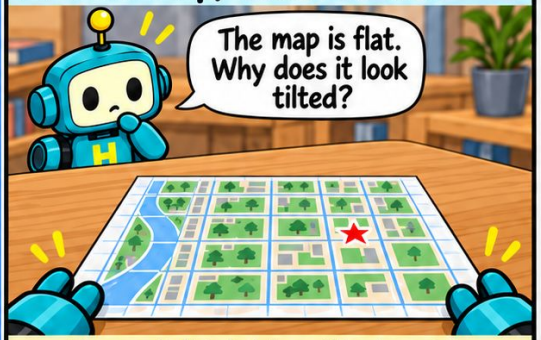
RECTIFICATION

Observe → **Match** → **Transform**

A MAP, TWO VIEWS

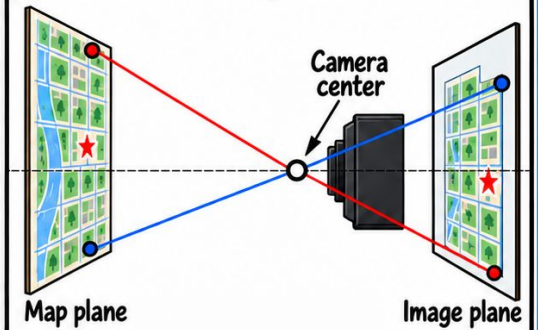
1 / Why does a flat map look skewed?

1. Same map, different view



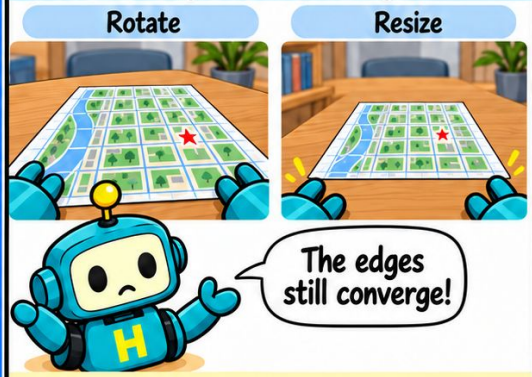
The map is flat, but from this viewpoint it looks tilted.

2. Follow the rays



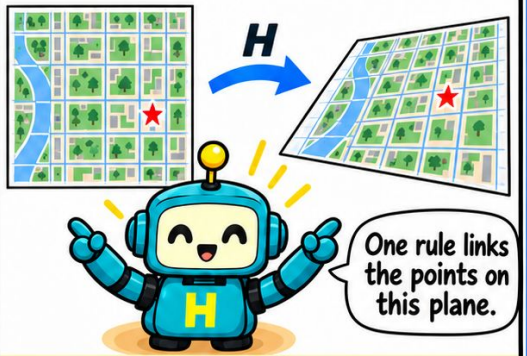
Light rays turn a 3D scene into a 2D image.

3. Rotating is not enough



Rotation and uniform scaling cannot remove perspective.

4. Meet the homography



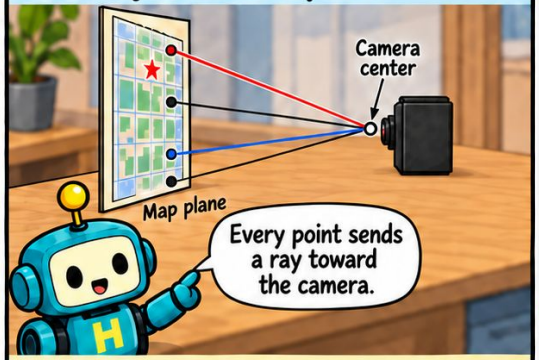
A homography maps one view of a plane to another.

PERSPECTIVE HOMOGRAPHY

FOLLOW THE LIGHT

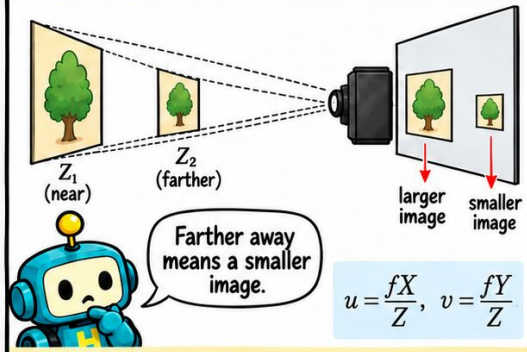
2 / How perspective forms

1. A ray for each point



A pinhole camera projects the scene onto an image plane.

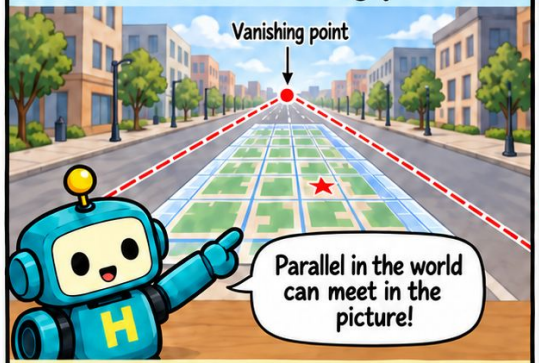
2. Divide by depth



$$u = \frac{fX}{Z}, \quad v = \frac{fY}{Z}$$

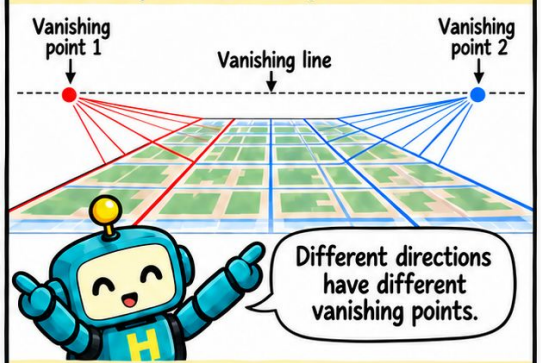
In this simplified model, depth Z appears in the denominator.

3. Meet the vanishing point



Lines parallel to the image plane may remain parallel.

4. One plane, many directions



For one plane, its vanishing points lie on a vanishing line.

PERSPECTIVE HOMOGRAPHY

ONE PLANE, TWO VIEWS

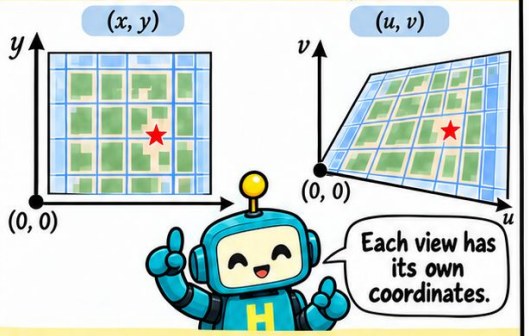
3 / What does a homography connect?

1. The same landmark



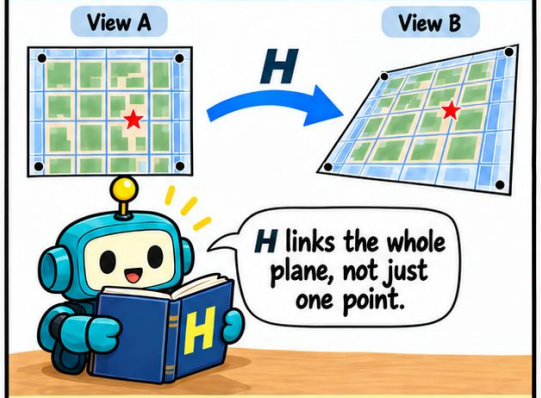
Corresponding points represent the same location on the plane.

2. Two coordinate systems



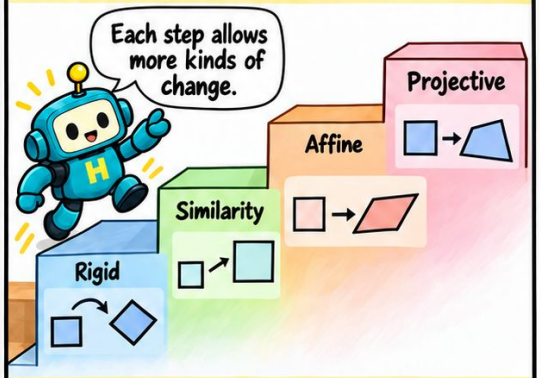
The source and target positions need not have the same numbers.

3. One rule for the plane



A homography is an invertible projective mapping of a plane.

4. A ladder of transformations



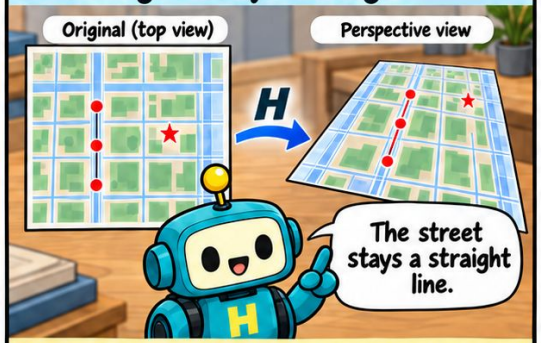
Rigid: move and rotate. Similarity: scale too.
Affine: keep parallels. Projective: allow convergence.

PERSPECTIVE HOMOGRAPHY

WHAT CHANGES? WHAT STAYS?

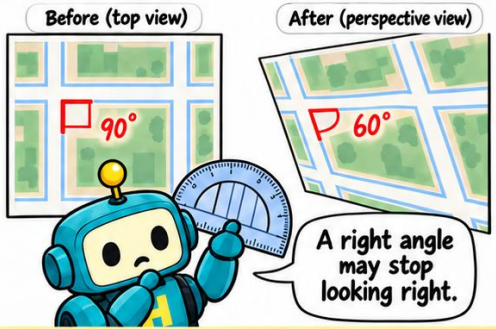
4 / The rules of projective geometry

1. Straight stays straight



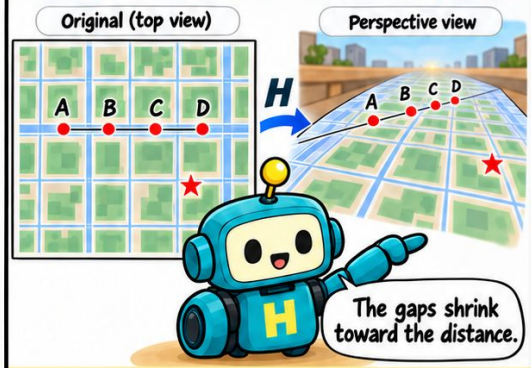
Homographies preserve straight lines, collinearity, and incidence.

2. Angles can change



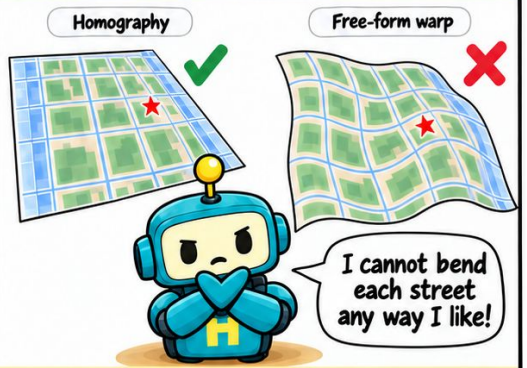
Angles, lengths, areas, and parallelism are not generally preserved.

3. Equal steps can look unequal



Midpoints and ordinary distance ratios need not stay the same.

4. Not a rubber sheet



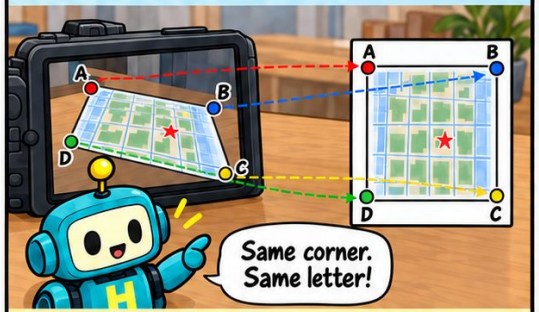
A single H imposes one consistent projective rule on the plane.

PERSPECTIVE HOMOGRAPHY

FOUR PAIRS, ONE RULE

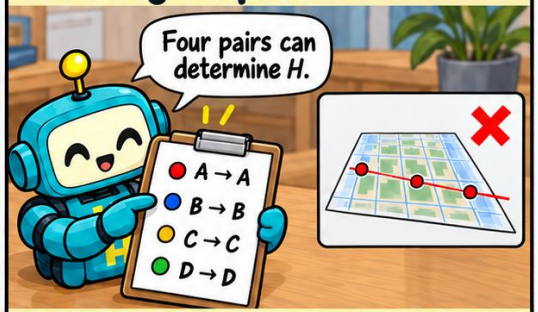
5 / Finding the mapping

1. Match the corners



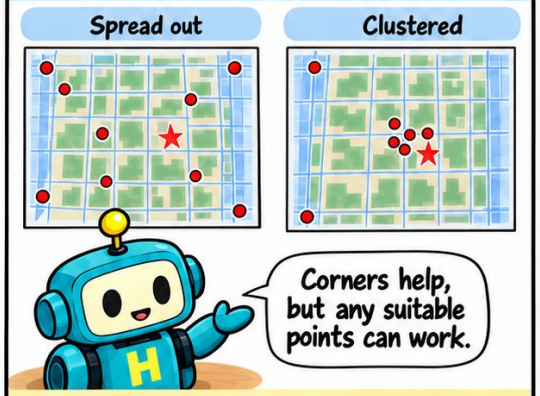
A point pair supplies two independent coordinate constraints.

2. Four good pairs



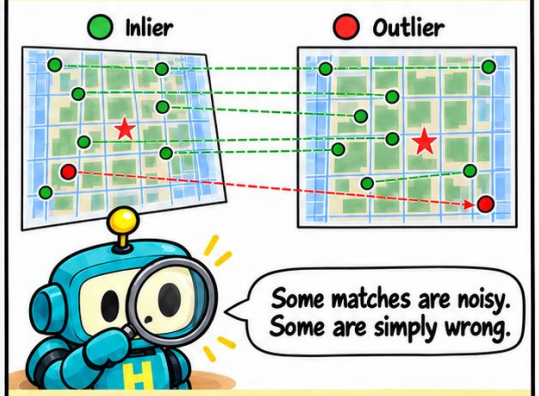
Use distinct points, with no three collinear in either set.

3. Spread out the evidence



Well-distributed correspondences usually give a more stable estimate.

4. Real matches are imperfect



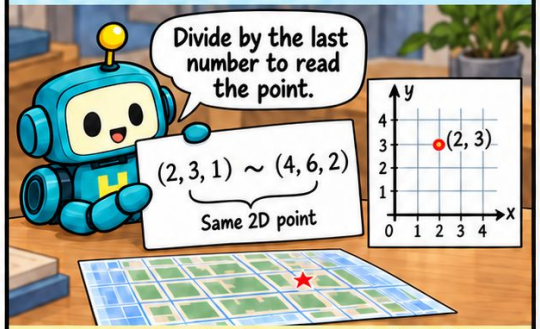
More matches help validation; RANSAC rejects inconsistent outliers. See page 9.

PERSPECTIVE HOMOGRAPHY

INSIDE THE MATRIX

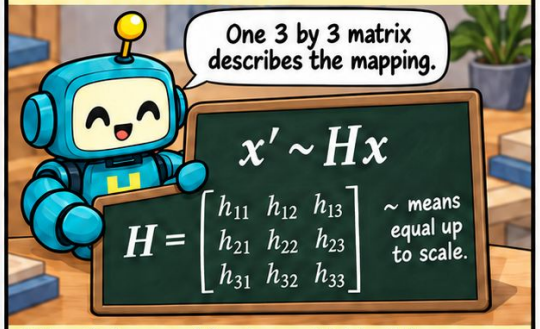
6 / Coordinates, H, and eight degrees of freedom

1. Add a coordinate



For $w \neq 0$, (a, b, w) represents $(a/w,\ b/w)$. The extra number is not depth.

2. Write the rule



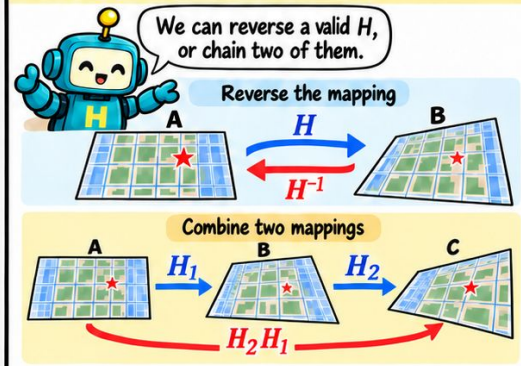
H has 8 degrees of freedom: multiplying it by a nonzero scale changes nothing.

3. The perspective division



If the last row is $(0, 0, 1)$, the mapping is affine.

4. Reverse and combine



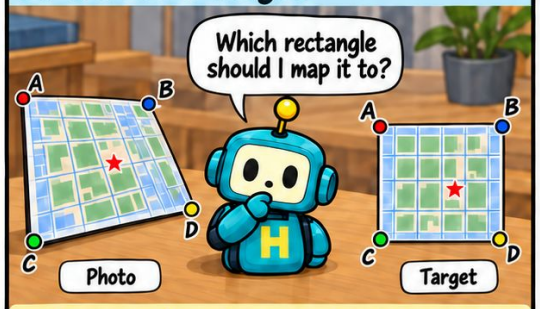
H must be invertible. The reverse uses H inverse; the combined map is $H_2 H_1$.

PERSPECTIVE HOMOGRAPHY

BRING THE MAP BACK

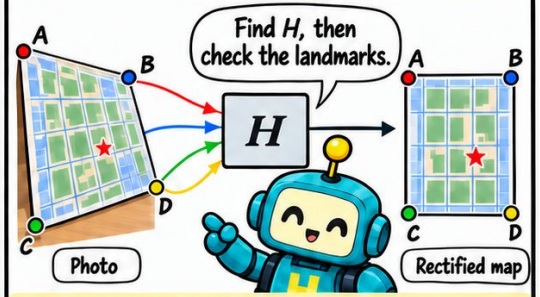
7 / Rectification and image resampling

1. Choose a target



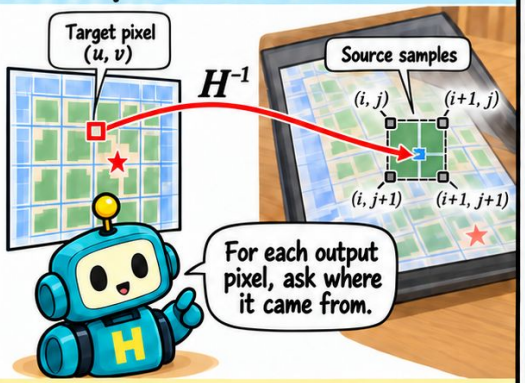
Use known dimensions or aspect ratio to recover meaningful proportions.

2. Estimate the mapping



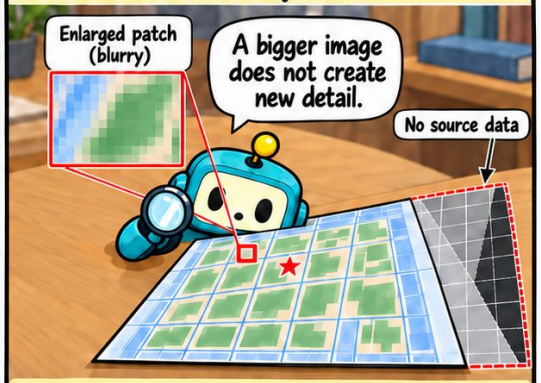
Turning a quadrilateral into a rectangle alone does not reveal its true aspect ratio.

3. Sample backward



Inverse mapping avoids scatter holes; interpolation estimates colors between samples.

4. Read the fine print



Resampling can soften details. Cropping and boundaries matter; hidden content stays missing.

PERSPECTIVE HOMOGRAPHY

POWERFUL, WITH LIMITS

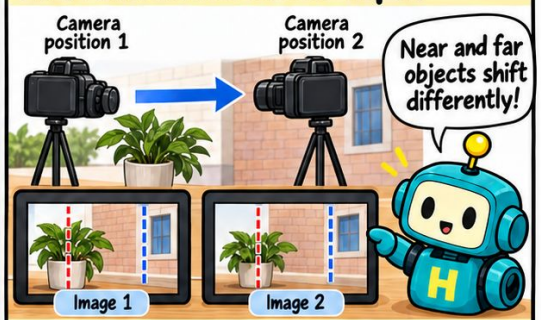
8 / Assumptions, failures, and applications

1. Flat works; curved does not



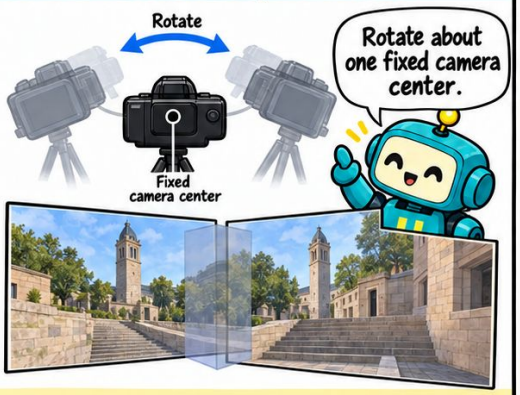
A curved surface usually needs a richer model than a single homography.

2. Translation reveals depth



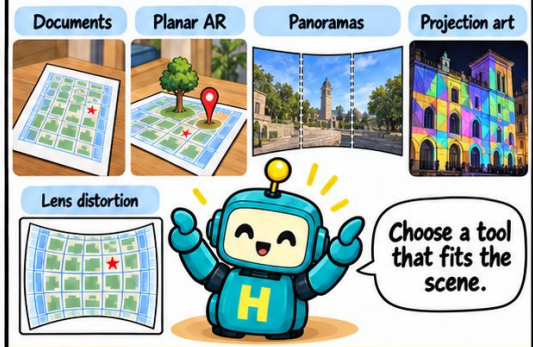
Camera translation creates parallax; one H usually cannot align all depths.

3. Rotation is a special case



Under ideal pinhole imaging, pure rotation also gives a homography.

4. Put it to work



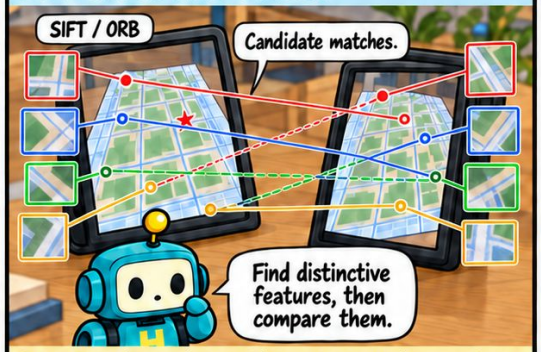
Homographies support alignment and planar effects.
Lens distortion needs separate correction.

PERSPECTIVE HOMOGRAPHY

WHEN MATCHES GET MESSY

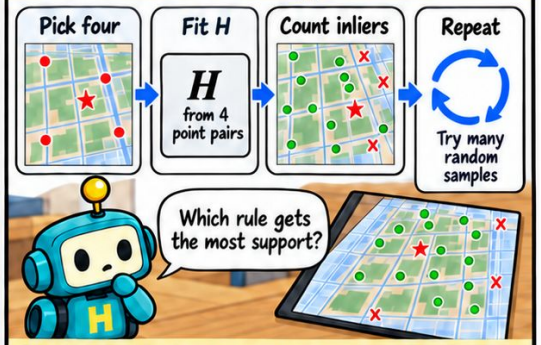
9 / Advanced: RANSAC, DLT, and refinement

1. Detect and match



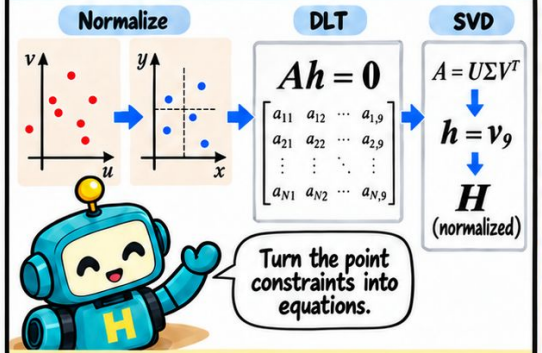
Automatic matching produces candidates, not guaranteed correspondences.

2. RANSAC tests hypotheses



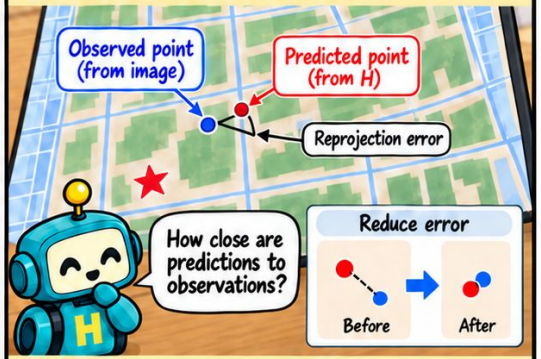
RANSAC tests minimal samples, scores consensus, then refits using inliers.

3. Solve with DLT



Normalize coordinates, solve $Ah = 0$ with SVD, reshape, then undo normalization to recover H.

4. Refine and validate



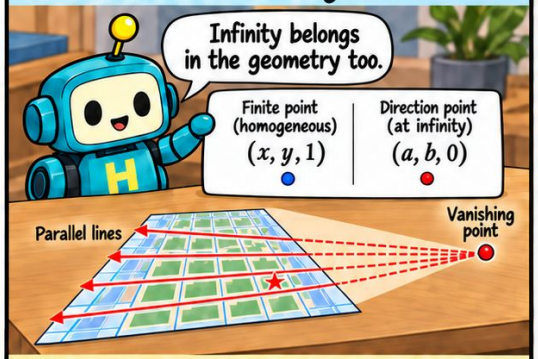
Refine geometric error and test held-out landmarks.
Low error cannot fix a wrong scene model.

PERSPECTIVE HOMOGRAPHY

BEYOND THE FLAT MAP

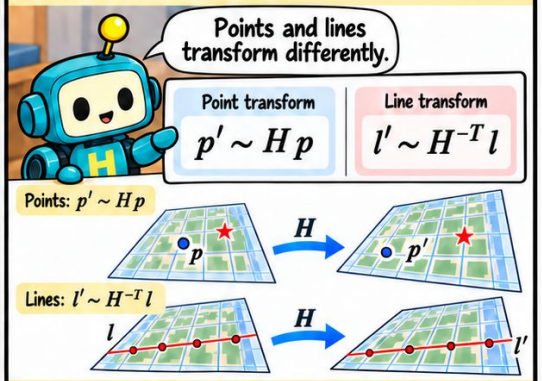
10 / Advanced: projective geometry and cameras

1. A home for infinity



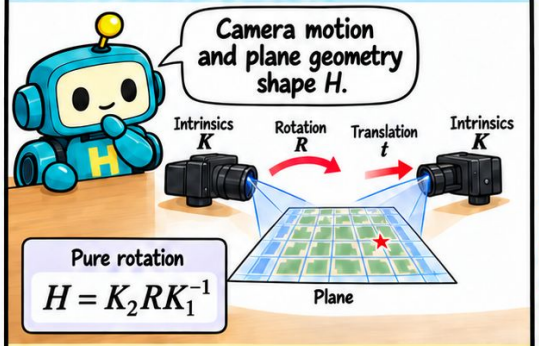
The projective plane includes points at infinity. Point–line incidence and cross-ratio are preserved.

2. Points and lines



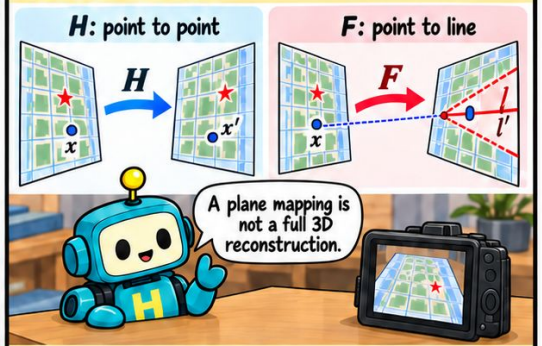
Four collinear points keep their cross-ratio, even when their gaps change.

3. A camera inside H



With calibration, decomposition may recover pose candidates; extra constraints resolve ambiguities.

4. H versus F



H maps plane points. F constrains general two-view matches to epipolar lines.